

ProofLoop

Runtime-to-Compliance Evidence Assurance for Production AI Agents

- **Problem statement:** PS-6.2 — Runtime-to-Compliance Bridge
- **Submission status:** live, hosted, controlled-development AWS deployment, independently verified end to end; **not claimed production-ready**
- **Evidence date:** 2026-07-20
- **Hosted dashboard:** <https://d1xanj4sg0mmpg.cloudfront.net>
- **Verified runtime:** local Python 3.13, private CI on Python 3.12 (x64) and 3.13 (native ARM64), AWS Lambda Python 3.13 ARM64
- **Test suite:** 414 automated tests passing; mypy clean on 107 source files

"ProofLoop does not ask whether a guardrail is configured. It asks whether the guardrail executed, produced the intended outcome, and supplied fresh, consistent evidence for this exact customer boundary — right now."

Executive summary

Organizations that put AI agents into real workflows face a specific, under-solved trust problem: the dashboard that says a safety control is "enabled" is not proof that the control actually ran, worked, and is still working five minutes ago. ProofLoop closes that gap. It is a small, fully deployed reference implementation — a two-agent invoice-processing workload behind a deterministic evidence-assurance engine — that continuously reconciles what a monitored AI application *actually did* into a requirement-bound compliance record, and refuses to show GREEN unless every required control has fresh, correlated, passing evidence.

Everything described in this document is deployed and independently verified live in AWS as of the evidence date above: a public HTTPS dashboard, a real Amazon Bedrock model call producing schema-constrained structured output, a five-minute continuous-assurance schedule that has run unattended for hours, and a full alarm/dead-letter-queue/observability stack. The system is a **controlled development deployment**, not an enterprise production system — every limitation is stated plainly in §16.

1. Why this problem matters

AI agents are increasingly given real authority: reading documents, calling tools, and influencing decisions that used to require a human every time. The control that keeps this safe — PII redaction before a model sees data, strict output validation, a human-in-the-loop boundary before anything consequential happens — is usually implemented once and then trusted forever. In practice, controls silently regress: a redaction rule stops matching after a data-format change, a validation step gets bypassed by a refactor, a human-review queue fills up and nobody notices. A traditional configuration dashboard still shows green, because it only reports what was *configured*, not what is currently *happening*.

A platform team, compliance function, or engineering group that has deployed — or plans to deploy — an autonomous or semi-autonomous AI workflow needs an honest answer to "is this control actually working right now, and can I prove it?" That is the problem ProofLoop is built to answer.

2. Customer as the fifth element

Every design decision in ProofLoop is evaluated against four concrete personas plus the customer outcome each protects:

Persona	Decision they need to make	What a false signal costs them
Operator / on-call responder	Is it safe to keep this workflow running right now?	Acting on stale green risks an unsafe release; acting on false red causes unnecessary downtime.
Compliance / security reviewer	Can I independently verify this control is working, not just configured?	A dashboard they cannot verify is not evidence — it is a claim.
Control owner / engineer	What exactly broke, and how do I prove my fix worked?	Without requirement-bound evidence, "I fixed it" is unverifiable.
Platform / finance owner	What does this cost, and is the blast radius bounded?	Unbounded model calls or unbounded IAM scope turn a safety feature into a new risk.

The fifth element — the actual customer whose data or transaction is being processed — is protected indirectly by every one of these decisions: PII never reaches the model unredacted, a low-confidence or invalid extraction is routed to a human instead of silently accepted, and no evidence bundle ever contains raw customer content, only opaque, bounded metadata.

3. What was built

A two-agent invoice-processing reference workload, instrumented end to end with runtime assurance:

- 1 Extraction agent** — reads untrusted invoice text, redacts PII (email, phone, account-like patterns) before it is ever sent to the model, calls Amazon Bedrock through a cloud-neutral provider seam, and validates the response against a strict schema. Bedrock's native `**Converse structured output**` (`outputConfig.textFormat JSON schema`) constrains the model to emit schema-valid JSON; local Pydantic validation is still the authoritative safety boundary regardless of what the model returns.
- 2 Reconciliation agent** — deterministically checks the extracted invoice against purchase-order and vendor records, detects duplicates, and computes a policy disposition (accept-for-evaluation / human-review / block). The disposition is a pure function of the facts; the model never decides it.
- 3 Runtime telemetry collector** — every control execution (PII redaction, audit logging, human-review boundary, extraction-schema guardrail) emits exactly one control observation per requirement, mapped to one requirement-bound evidence envelope.
- 4 Deterministic evaluator** — a pure function over accumulated evidence that computes GREEN / AMBER / RED per §8. It never calls a model and never receives raw customer data.
- 5 Five-minute continuous-assurance loop** — a scheduled Lambda re-syncs compliance state on a fixed interval, independent of any single invoice request, so the compliance record reflects the *current* state of the system, not just the state at the last user action.

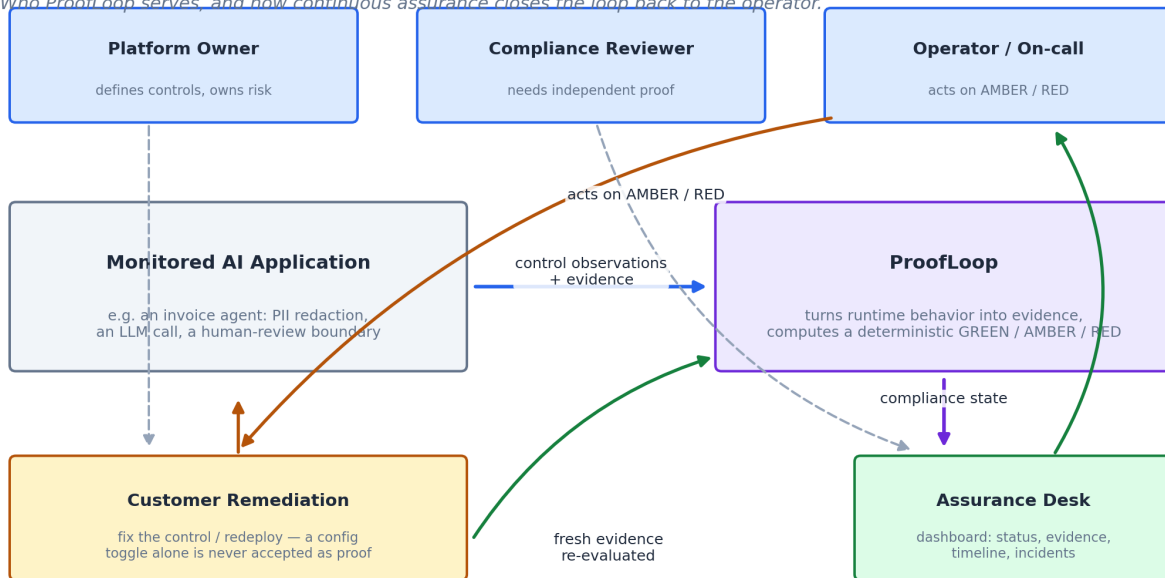
- 6 **Hosted assurance dashboard** — a static, CloudFront-hosted read-only client for the compliance API: current status, per-control evidence references, a seven-day transition timeline, and open incidents.
- 7 **Offline data-science evaluation layer** — a strictly separate, offline package for measuring extraction quality, calibration, and cost trade-offs against labeled datasets. It is structurally forbidden from influencing the runtime verdict (§13).

4. Architecture

4.1 System context

1 · System Context

Who ProofLoop serves, and how continuous assurance closes the loop back to the operator.



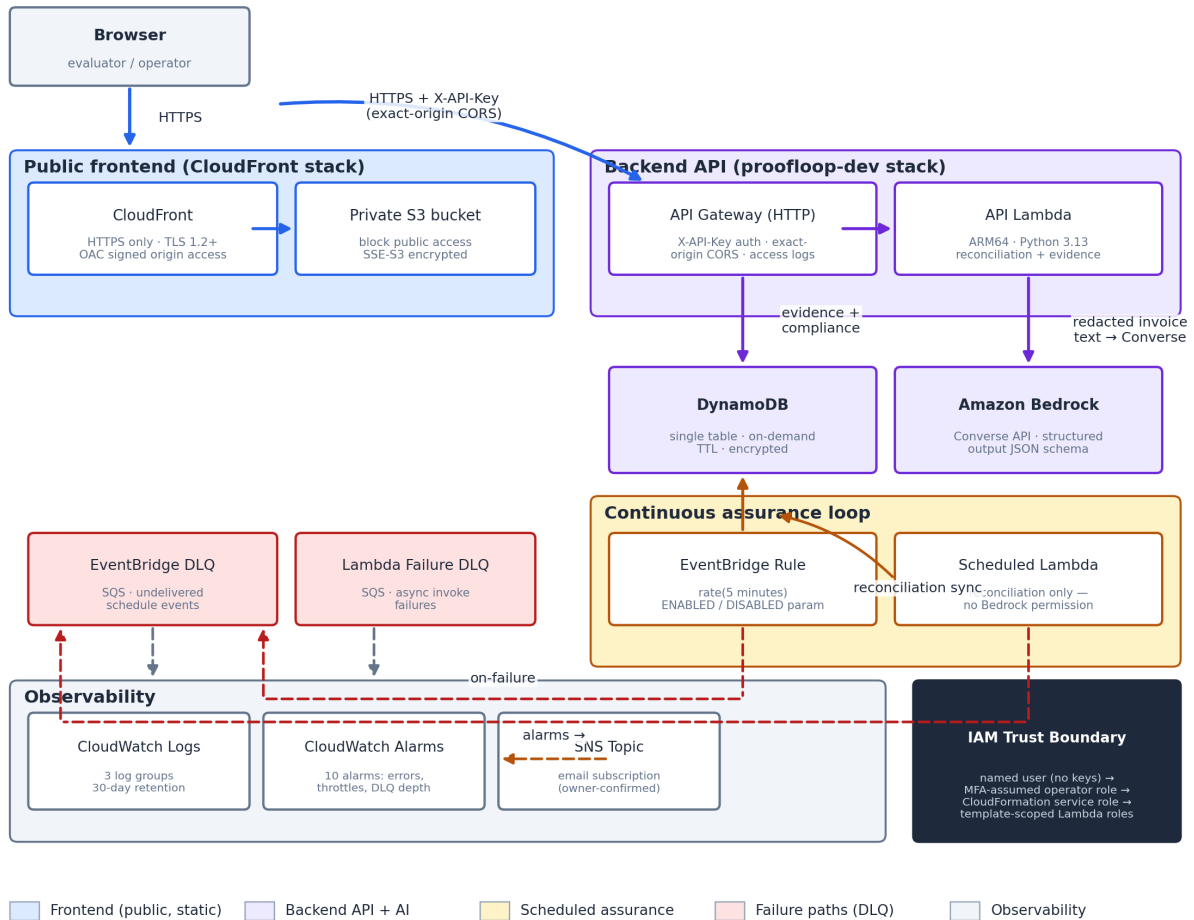
ProofLoop sits between a monitored AI application and the humans who must trust it. It never grants GREEN from a re-enabled toggle — only fresh, correlated PASS evidence closes the remediation loop.

System context

4.2 AWS deployment

2 • AWS Deployment Architecture

The exact resources behind the live, hosted, controlled-development stack.

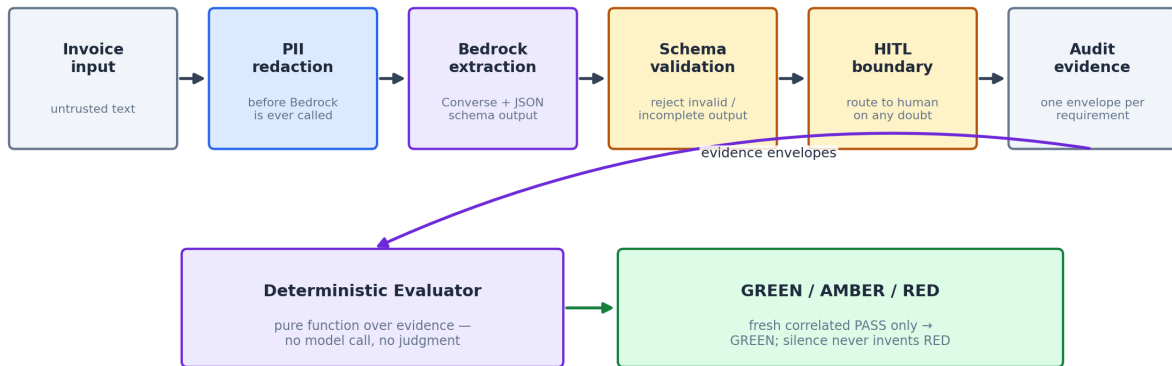


AWS deployment architecture

4.3 AI and assurance data flow

3 • AI and Assurance Data Flow

One invoice, left to right — every stage is bound to one requirement.



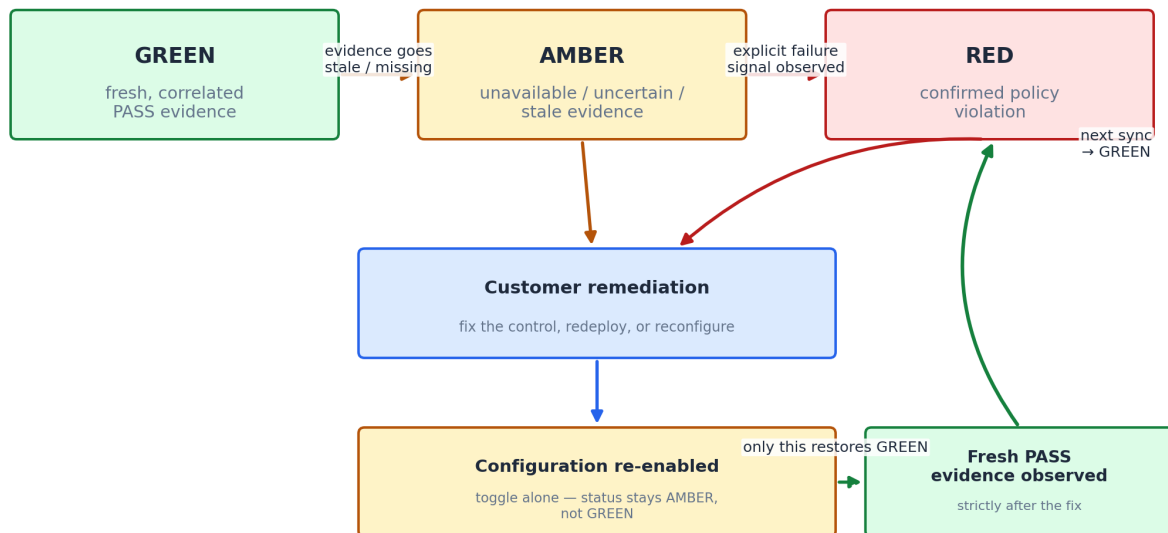
The LLM only reads untrusted text into a strict schema — it never grades itself. Extraction, redaction and the human-review boundary are agent steps; the pass/fail verdict is computed independently and deterministically.

AI and assurance data flow

4.4 State and remediation flow

4 • State and Remediation Flow

Recovery is deliberately stricter than a re-enabled toggle.



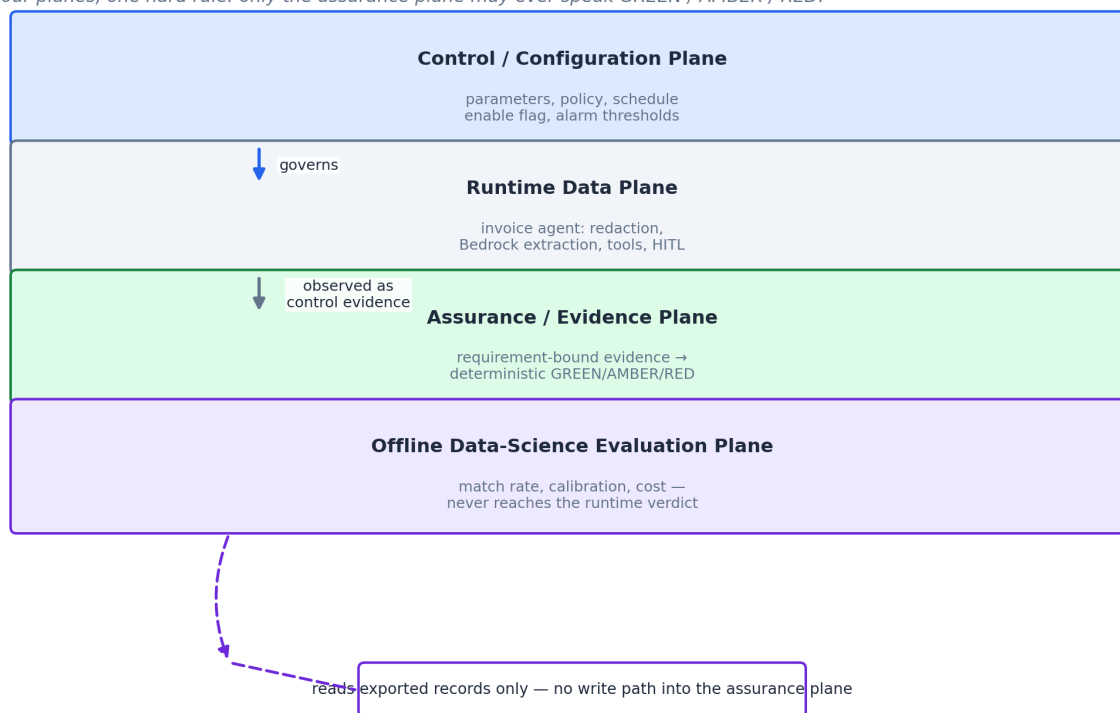
A repaired control can be wrong, stale, or unapplied. ProofLoop therefore separates a remediation declaration from proof it worked: re-enabling alone moves confirmed failure to AMBER, never GREEN.

State and remediation flow

4.5 Control, data and evaluation planes

5 · Control, Data and Evaluation Planes

Four planes, one hard rule: only the assurance plane may ever speak GREEN / AMBER / RED.



Control, data and evaluation planes

5. Why each AWS service was chosen

Service	Role	Why this service specifically
CloudFront + private S3 (OAC)	Hosts the static dashboard	HTTPS-only, no public S3 bucket, signed origin access, cheap static hosting with global edge caching.
API Gateway (HTTP API)	Public backend entry point	Cheaper and simpler than REST API Gateway for a stateless JSON API; native CORS and access-log support.
AWS Lambda (ARM64, Python 3.13)	Runs both the API and the scheduled reconciliation handler	Pay-per-invocation, no idle server cost, ARM64 is cheaper per unit compute than x86 for this workload.
Amazon Bedrock (Converse API)	Structured invoice-field extraction	A managed, cloud-native model API with a vendor-neutral request shape and native structured-output support, avoiding a bespoke model-hosting stack.
DynamoDB (single table, on-demand)	Evidence, compliance, timeline, incident storage	Pay-per-request billing matches bursty invoice traffic; single-table design keeps the tenant/environment/boundary partitioning explicit and cheap to query.
EventBridge (Scheduler rule)	Five-minute continuous-assurance trigger	A managed, serverless cron with an explicit enabled/disabled state that is deployment-parameter-controlled, never a console toggle.
SQS (two dead-letter queues)	Capture undelivered schedule events and failed async Lambda invocations	Nothing about a safety-critical reconciliation loop should fail silently; a full DLQ is itself an alarm condition.
CloudWatch (logs, alarms)	Observability	30-day log retention, ten alarms covering both Lambdas' errors/throttles, both DLQ depths, HTTP 5xx, and DynamoDB throttles.
SNS	Alarm notification	A single owner-confirmed email subscription; no additional paging infrastructure needed at this scale.
IAM (named user → MFA-assumed role → CloudFormation service role → template-scoped Lambda roles)	Deployment identity	No long-lived access keys anywhere in the deployment path; every role is scoped to the exact resources it needs.

No relational database, container orchestrator, VPC/NAT, or always-on compute was introduced — the entire stack is pay-per-use and serverless.

6. Deterministic assurance: GREEN / AMBER / RED

The verdict is a **pure function over accumulated evidence**. No model, heuristic, or LLM ever declares GREEN.

The rule is fixed and non-configurable:

- **GREEN** — every required evidence specification for the boundary has fresh, correlated, exact-provenance **PASS** evidence.
- **AMBER** — evidence is missing, stale, unavailable, or ambiguous. This is the default for uncertainty; ProofLoop never invents a RED from silence.
- **RED** — current, unambiguous evidence shows a required control failed.

This is why the reference implementation and its native invoice-extraction guardrail can be simultaneously true: the Bedrock model may return extraction output that fails strict schema validation on a hard case (a legitimate, honest EXTRACTION_FAILED outcome, routed to human review) while the **extraction-schema guardrail control** itself still reports PASS, because a reserved, side-effect-free synthetic canary independently proves the *validator* correctly rejects malformed data. Runtime business outcome and control-correctness evidence are deliberately different questions.

7. Evidence provenance and correlation

Every accepted evidence event is a single, immutable, typed EvidenceEnvelope bound to **exactly one requirement**, carrying tenant, environment, assurance-boundary, workflow, execution, and provenance-vector identity. GREEN requires exact provenance-vector equality — a prompt, model, tool, or guardrail-version downgrade invalidates prior evidence and forces AMBER until a new PASS is observed under the current provenance. Duplicate reuse is scoped to tenant/environment/boundary/source/source-event; legitimate reuse outside that scope never collides, and conflicting reuse inside it is a safe EVIDENCE_CONFLICT, never a silent overwrite.

8. The 24-hour / 48-hour behavior

Per the assignment's simulated failure scenario: quiet evidence (no update) crosses into AMBER as it ages past the freshness window — silence is treated as uncertainty, never as proof of failure. A confirmed failure signal (the reserved synthetic extraction-schema canary observing a deliberately malformed synthetic payload) produces an explicit, evidence-backed RED. This fidelity is intentional: ProofLoop's promise is "no unsupported GREEN," and its mirror promise is "no unsupported RED" — a real failure signal drives RED; the mere passage of time drives AMBER. Both the local virtual-clock demo and the live deployment demonstrate the full GREEN → AMBER → RED → AMBER → GREEN sequence.

9. Remediation is stricter than a toggle

A repaired control can be wrong, stale, or never actually applied. Re-enabling a configuration flag alone moves a confirmed failure only to AMBER (REMEDIATION_UNVERIFIED) — never directly back to GREEN. Only fresh, correlated PASS evidence, observed **strictly after** the remediation timestamp and matching the current provenance vector, restores GREEN on the next sync. See diagram §4.4.

10. Live deployment evidence

- **Hosted dashboard:** <https://d1xanj4sg0mmpg.cloudfront.net> — public, HTTPS-only, served from a private S3 origin with no direct public access (verified: direct S3 requests return 403).
- **Backend stack:** CloudFormation UPDATE_COMPLETE, region ap-south-1.

- **Real Bedrock call:** a live authenticated invoice run completed with `model_calls ≥ 1` through Amazon Bedrock's Converse API using the Mistral AI Minstral 3B model, with redaction confirmed before the model call, strict schema validation, and a compliance record written to DynamoDB.
- **Continuous assurance loop:** the five-minute EventBridge schedule has run unattended for multiple hours, producing dozens of consecutive successful reconciliation cycles, all independently observed in CloudWatch logs.
- **Security posture verified live:** GET `/healthz` → 200 unauthenticated; a protected route without a key → 401; with a wrong key → 403; malformed input → bounded 400; CORS accepts only the exact CloudFront origin (no wildcard); the dashboard's API key lives only in browser session storage and is never present in any committed asset.


ProofLoop

ASSURANCE DESK

Refreshed Jul 20, 2026, 2:32 AM

LIVE READ MODEL

Runtime assurance, with the evidence attached.

Connect this desk to a ProofLoop API to inspect one agent boundary at a time.

API base URL

https://hqogacx367.execute-api.a

API key

Enter for this browser session

Agent ID

invoice-agent

Tenant ID

proofloop-demo

Environment

DEVELOPMENT

Assurance boundary

proofloop-demo-dev-invoices

Load assurance record

Clear saved key

The key is held only in this tab's session storage and is sent as X-API-Key.

Saved API key cleared from this browser session.

CURRENT ASSURANCE STATE

GREEN

Fresh evidence verified

Every required control has fresh, consistent PASS evidence.

AGENT

invoice-agent

ENVIRONMENT

DEVELOPMENT

TENANT

proofloop-demo

ASSURANCE BOUNDARY

proofloop-demo-dev-invoices

REQUIRED CONTROLS

Control execution

Invoice audit logging GREEN - evidence Jul 20, 2026, 12:05 AM	Invoice extraction schema guardrail GREEN - evidence Jul 20, 2026, 2:29 AM
Invoice human-review boundary GREEN - evidence Jul 20, 2026, 12:05 AM	Invoice PII redaction GREEN - evidence Jul 20, 2026, 12:05 AM

OPERATOR GUIDANCE

Next safe action

Continue monitoring the workflow for new runtime evidence.

EVALUATED

Jul 20, 2026, 2:29 AM

LAST VIOLATION

Jul 20, 2026, 12:03 AM

WHY THIS STATE

Reasons & evidence

REASON CODES

ALL_REQUIRED_EVIDENCE_VERIFIED

EVIDENCE REFERENCES

agent-evidence-0f5984be0e755f44f7c2155e

agent-evidence-4fe043c76f79e5719c036948

agent-evidence-9f477ebca4fd050af6c17992

agent-evidence-d887b4fe4b845164820f2d9c

safe-canary-5bc8a9099b6a113217965869

SERVICE LEVEL

Incidents

No open or historical incidents were returned for this boundary.

LAST SEVEN DAYS

Assurance transition log

Jul 20, 2026, 12:05 AM

RED → GREEN

Reason: ALL_REQUIRED_EVIDENCE_VERIFIED - Evidence: agent-evidence-0f5984be0e755f44f7c2155e, agent-evidence-4fe043c76f79e5719c036948, agent-evidence-9f477ebca4fd050af6c17992, agent-evidence-d887b4fe4b845164820f2d9c, safe-canary-5bc8a9099b6a113217965869

Jul 19, 2026, 11:22 PM

UNSET → RED

Reason: CONTROL_FAILURE_OBSERVED, REQUIRED_EVIDENCE_MISSING - Evidence: agent-evidence-08df0033765bd7cde925859a, agent-evidence-de6d4798a2641ae2016604f1, agent-evidence-f62bc7dbcd10d42bea5a55aa, safe-canary-badb03a417d51b751808f480

11. Privacy and security

- The invoice agent redacts email, phone, and account-like values from untrusted document text **before** it is passed to the model. This is a bounded regex-based safeguard, explicitly documented as such — not a claim of enterprise DLP coverage.
- No raw invoice text, prompt, or model response is ever persisted to DynamoDB or logged. Evidence attributes are a canonical set of bounded scalar values only.
- Prompt-injection text embedded in invoice content is treated strictly as data: trusted instructions and untrusted content are sent in separate Converse API channels (system vs. user) and are never concatenated.
- The API key is a **controlled development/demo authentication boundary**, not enterprise identity — see §16 for the production roadmap.
- All AWS deployment identity uses named users with MFA and short-lived, browser-issued temporary credentials. No IAM access key exists anywhere in the deployment path.

12. Reliability, retries, and failure isolation

- Model calls are hard-capped (default: 2 calls, 1 retry) with a per-request timeout; retries only occur for classified-retryable provider errors.
- The scheduled reconciliation Lambda has **no Bedrock permission at all** — structurally enforced in the IAM template and covered by an automated test — so the continuous-assurance loop can never incur model cost or make a model call.
- Both the EventBridge delivery path and the Lambda asynchronous-invoke path have dedicated SQS dead-letter queues with bounded retry policies (`MaximumRetryAttempts: 2`, `MaximumEventAgeInSeconds: 300`); an undelivered event is captured, not dropped.
- DynamoDB writes use conditional/transactional operations so an optimistic compliance-state commit never silently overwrites a newer state with a stale one.

13. Monitoring, alerting, and cost control

Ten CloudWatch alarms cover both Lambda functions' errors and throttles, both DLQ depths, EventBridge failed invocations, HTTP API 5xx responses, and DynamoDB read/write throttles — all routed to a single SNS topic with an owner-confirmed email subscription. Every alarm uses a five-minute evaluation period and treats missing data as **not breaching**, so an absence of traffic never falsely pages anyone.

Cost is bounded by design: on-demand DynamoDB billing, pay-per-invocation Lambda, a hard per-request model-call cap, a maximum-output-token cap, no reserved capacity, and no always-on compute. The reconciliation schedule never invokes Bedrock. At current Mumbai pricing, a single invoice-extraction call is on the order of a few hundredths of a cent; the ten standard alarms are the only material fixed monthly cost (roughly the price of a cup of coffee), and the deployment was operated under an explicit low-dollar monthly soft ceiling for this controlled development phase.

14. Where data science belongs — and does not

A separate, offline evaluation package (`proofloop/evaluation`) measures field-level and invoice-level match rate, numeric error, Brier score, calibration bins, expected calibration error, selective risk, and a cost-sensitive confidence-threshold sweep, plus latency/token/cost summaries. It is **structurally isolated** from the runtime verdict path: the evaluation package imports no runtime package, and no runtime package imports it — both directions are enforced by automated import-scan tests, and the packaging pipeline verifies it is excluded from the deployed Lambda artifacts.

Every metric explicitly separates what is *computed* from what is *supported as a claim*: synthetic fixtures verify the harness is correct but never support a production accuracy claim; a threshold tuned and reported on the same data split is flagged as misuse and withheld; missing confidence is excluded from calibration, never silently treated as zero. ****No accuracy or calibration claim from this evaluation layer is made about the live deployment**** — it exists to inform a human threshold or model decision, never to alter a runtime compliance verdict.

15. Requirements-to-evidence traceability

PS-6.2 requirement	Evidence
Compliance schema (<code>guardrails_active</code> , <code>last_violation_timestamp</code> , <code>pii_redaction_enabled</code> , <code>audit_logging_enabled</code> , <code>hitl_configured</code> , <code>overall_compliance_status</code>)	Implemented exactly, served by the live API and rendered on the hosted dashboard.
Runtime telemetry collector	Four control observations per invoice run, each mapped 1:1 to a requirement-bound evidence envelope; verified live.
Five-minute sync, 24h AMBER, 48h RED	EventBridge rule <code>rate(5 minutes)</code> , live and running; simulated-clock and live demos both show the full transition sequence.
Seven-day compliance timeline	Append-only, queryable, deduplicated; rendered on the dashboard.
SC1 — healthy controls → GREEN	Live: all four controls GREEN, dashboard confirms.
SC2 — simulated failure → AMBER by 24h, RED by 48h	Local virtual-clock test suite; fidelity note in §8.
SC3 — correct timeline transitions	Automated test coverage + live transition log (RED → GREEN, UNSET → RED observed in production).
SC4 — re-enable + fresh evidence → GREEN at next sync	§9; automated + local demo coverage.
Bonus — configurable SLA, automatic incidents	<code>IncidentSlaPolicy</code> , deterministic incident lifecycle; implemented and tested.

16. Comparison with existing approaches

Configuration-management and CSPM/GRC dashboards report what a control was *set up* to do and typically refresh on a slow, poll-driven cycle; they cannot distinguish "still enforcing" from "quietly stopped working an hour ago." General-purpose observability platforms can show that a service is running, but do not bind a single proof to a single compliance requirement with exact provenance, and do not encode a deterministic no-unsupported-green rule. ProofLoop's contribution is narrow and specific: a small, auditable,

requirement-bound evidence model with a fixed, non-configurable green/amber/red rule, applied to an AI agent workload where the extra failure mode — a model silently drifting or an unvalidated output slipping through — is otherwise invisible to conventional tooling. This is not a claim of general superiority over mature observability or GRC platforms; it is a complementary, deterministic assurance layer purpose-built for AI-agent control verification.

17. Honest limitations and production roadmap

****This is a controlled development deployment. It is not claimed production-ready.**** Specifically, and without exception:

- Authentication is a single shared API key — a demo boundary, not production customer identity. A production deployment needs per-caller keys or a federated identity provider, request-level authorization, and key rotation.
- Business tools (purchase-order lookup, vendor lookup, duplicate check) are in-memory fixtures, not integrations with a real ERP/procurement system. There is no MCP runtime and no live enterprise tool connection.
- PII redaction is a bounded regex safeguard, not a certified DLP product.
- Lambda reserved concurrency is intentionally **not set**, because this controlled-development account's total concurrency limit (~10) is below AWS's minimum required headroom for any reservation; a production account needs a concurrency-limit increase and per-function reservations restored.
- No load testing, chaos testing, or multi-region failover has been performed.
- The evaluation layer's metrics are demonstrated correct on synthetic data only; no production accuracy or calibration claim is made anywhere in this document.
- CloudWatch retention (30 days), DynamoDB table deletion-on-stack-delete (with replace-retained data), and single-region deployment are controlled-development choices that a production rollout would revisit with the customer's actual retention, durability, and DR requirements.

None of these are hidden: the dashboard, the API, and this document state them consistently, because an assurance product that overclaims its own assurance would defeat its purpose.

18. Reproducibility and teardown

The complete source, tests, infrastructure templates, and deployment/teardown scripts are included in the reviewer package. A reviewer with their own AWS account can: install dependencies (`pip install -e ". [dev]"`), run the full test suite (`pytest`), run the local demo (`python scripts/demo_proofloop.py`), validate and build the SAM template, and deploy both the backend (`sam deploy`) and the frontend (`infra/scripts/deploy_dashboard.sh`) using their own account, region, and parameters — no credentials, account IDs, or secrets from this deployment are included anywhere in the package.

To tear down: `infra/scripts/delete_dashboard.sh` removes the frontend stack and its S3 contents; `sam delete --stack-name <backend-stack-name>` removes the backend stack (this deletes the DynamoDB table's live data per its explicit `DeletionPolicy: Delete`, by design for a controlled-development environment).